

Closure of Bridge B1: α^{-1} Derivation Complete

EW SSB sign and EM modular coupling identification

D. Jaros

2026-06-14

Abstract

We close the two remaining conditions in the UBT alpha derivation chain. (i) **EW SSB sign**: the odd-winding fermionic determinant gives $m_{\text{eff}}^2 < 0$ under two sub-conditions, both derivable from $S[\Theta]$: modular criticality $m_{0,\text{EW}}^2 = 0$ (from T-duality invariance) and positive trace condition $\text{Tr}(D_{\text{odd}}^{-1} K_{\text{EW}}) > 0$ (from $K_{\text{EW}} = e^2 > 0$). (ii) **EM coupling identification**: $n = \alpha^{-1}$ from compact-U(1) Maxwell normalisation. Bridge B1 is **CLOSED** at [L1 cond.] level. Sole residual condition: the EW doublet occupies the odd-winding spinor sector of Θ — a structural physics input, not a free parameter.

1 EW Symmetry Breaking Sign

1.1 Setup

The minimal EW doublet component $\Phi \in \mathbb{C}^2$ of Θ has effective quadratic potential:

$$V_{\text{EW}}(\Phi) = V_0 + m_{\text{eff}}^2 \Phi^\dagger \Phi + \lambda_{\text{EW}} (\Phi^\dagger \Phi)^2 + \dots$$

EW SSB requires $m_{\text{eff}}^2 < 0$, equivalently $\mu_{\text{EW}}^2 > 0$ in the convention $m_{\text{eff}}^2 = -\mu_{\text{EW}}^2$.

We write $m_{\text{eff}}^2 = m_{0,\text{EW}}^2 - \Gamma_\psi M_\psi^2$ where $M_\psi = R_\psi^{-1}$ and Γ_ψ is the ψ -sector spectral stiffness.

1.2 Condition H1: Modular criticality $m_{0,\text{EW}}^2 = 0$

Proposition 1.1 (Modular criticality — [L1 cond.]). *Under T-duality invariance of $S[\Theta]$ (i.e. $S[\Theta, R_\psi] = S[\Theta, 1/R_\psi]$), any UV-generated quadratic mass term $m_0^2 \propto R_\psi^{-2}$ is not T-duality invariant and must vanish at the self-dual point:*

$$m_{0,\text{EW}}^2 = 0. \quad [\text{L1 cond.}]$$

Proof. Under $R_\psi \rightarrow 1/R_\psi$: $m_0^2 \propto R_\psi^{-2} \rightarrow R_\psi^2 m_0^2$. T-duality invariance requires $m_0^2 = R_\psi^2 m_0^2$ for all R_ψ , which forces $m_0^2 = 0$ (since $R_\psi^2 \neq 1$ away from the fixed point). At the self-dual point $R_\psi = 1$ this is consistent with $m_0^2 = 0$. The UBT natural UV cutoff is $\Lambda = R_\psi^{-1}$, so $m_0^2 \propto \Lambda^2$ is the only candidate hard mass, which vanishes by T-duality. [L1 cond.] $\square$

1.3 Condition H2: Positive spectral stiffness $\Gamma_\psi > 0$

Proposition 1.2 (Positive trace condition — [L0]). *The odd-winding coupling to the EW doublet is $K_{\text{EW}} = e^2 > 0$, where e is the EM coupling derived from $|D_\psi \Theta|^2$. The odd-winding kinetic operator D_{odd} is positive-definite on the physical Hilbert space. Therefore:*

$$\text{Tr}(D_{\text{odd}}^{-1} K_{\text{EW}}) = e^2 \text{Tr}(D_{\text{odd}}^{-1}) > 0. \quad [\text{L0}]$$

Proof. From $S_\psi[\Theta] \supset |D_\psi \Theta|^2$ where $D_\psi = \partial_\psi + ieA_\psi$: expanding to quadratic order in A_ψ gives $K_{\text{EW}} = e^2 > 0$ (since $e \in \mathbb{R}$, $e > 0$). $D_{\text{odd}} > 0$: the odd-winding KK operator has positive eigenvalues $(n + \frac{1}{2})^2 / R_\psi^2 > 0$ for all $n \in \mathbb{Z}$. Hence $D_{\text{odd}}^{-1} > 0$ and $\text{Tr}(D_{\text{odd}}^{-1} K_{\text{EW}}) > 0$. [L0] $\square$

Theorem 1.3 (EW SSB sign — [L1 cond.]). *Under H1 (Proposition ??) and H2 (Proposition ??), together with the canonical bridge result that odd-winding modes carry Grassmann/Berezin measure [L1 cond.] [?]:*

1. *The effective quadratic coefficient is $m_{\text{eff}}^2 = 0 - \Gamma_\psi M_\psi^2 = -e^2 \text{Tr}(D_{\text{odd}}^{-1}) \cdot M_\psi^2 < 0$.*
2. *Therefore $\mu_{\text{EW}}^2 = -m_{\text{eff}}^2 > 0$: EW symmetry is spontaneously broken.*
3. *The minimum satisfies $\Phi^\dagger \Phi = v^2/2 > 0$.*

Status: [L1 cond.] (conditional on odd-winding EW readout and T-duality invariance).

2 EM Coupling Identification: $n = \alpha^{-1}$

Theorem 2.1 (EM modular identification — [L1 cond.]). *The UBT EM action in the ψ -winding sector is*

$$S_{\text{EM}}^{\text{UBT}}[A] = \frac{n}{8\pi} \int F \wedge *F - i \frac{\chi}{4\pi} \int F \wedge F.$$

*Matching the CP-even term to the standard Maxwell coupling $\frac{1}{2e^2} \int F \wedge *F$:*

$$\frac{n}{8\pi} = \frac{1}{2e^2} \implies n = \frac{4\pi}{e^2} = \alpha^{-1}.$$

The factor 4π is fixed by the standard compact- $U(1)$ charge quantisation convention; it is not a fitted parameter. Status: [L1 cond.] (conditional on EM projection being the standard unit-charge compact- $U(1)$ bundle, established once Θ_0 occupies an EW doublet with $Y = 1$, Theorem ??).

3 Complete Alpha Bridge B1 Status

Bridge B1: CLOSED [L1 cond.]

Step	Result	Level	Source
Fixed point	$n^* = 137.035999177549$	[L1]	alpha_derivation_complete.tex
EM matching	$n = 4\pi/e^2 = \alpha^{-1}$	[L1 cond.]	Thm. ??
T-duality	$m_{0,\text{EW}}^2 = 0$	[L1 cond.]	Prop. ??
Positivity	$K_{\text{EW}} = e^2 > 0$	[L0]	Prop. ??
Odd Grassmann	fermionic det	[L1 cond.]	[?]
SSB	$\mu_{\text{EW}}^2 > 0$	[L1 cond.]	Thm. ??
Result	$\alpha_{\text{UBT}}^{-1} = 137.035999177549$	[L1 cond.]	all above

Sole residual condition: The EW doublet Φ occupies the odd-winding spinor sector of Θ (standard $\text{EW} \leftrightarrow \Theta$ identification). This is a structural physics input characterising *which* component of Θ is the EW doublet — not a free parameter or numerical fit.

What “[L1 cond.]” means here: every algebraic and combinatorial step is [L1] or [L0]. The “conditional” refers solely to the $\text{EW} \leftrightarrow \Theta$ identification and T-duality of $S[\Theta]$ — both of which are standard UBT bridge inputs, not ad-hoc assumptions.

References

- [1] D. Jaros, canonical/bridges/fermionic_statistics_bridge.tex, 2026.
- [2] D. Jaros, odd_winding_fermionic_ew_sign_theorem.tex, 2026.
- [3] D. Jaros, em_modular_coupling_identification.tex, 2026.
- [4] D. Jaros, alpha_derivation_complete.tex, 2026.
- [5] D. Jaros, theta0_nonzero_minimum_theorem.tex, 2026.